

Fear of the invader: a predatory invader may provoke stronger non-consumptive effects on a keystone native prey than a native predator

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Non-consumptive effects (NCEs) of predators on prey encompass costly antipredator behaviours such as increased refuge use and reduced foraging, which can influence prey population dynamics and community structure, particularly when the prey themselves are keystone species. In Aotearoa-New Zealand, the native freshwater crayfish kōura (*Paraneophrops planifrons*) is a keystone species, an ecosystem engineer with a critical functional role in sediment dynamics, nutrient cycling and benthic community structure. Kōura are preyed upon by both the co-evolved native longfin eel (*Anguilla dieffenbachii*) and invasive brown bullhead catfish (*Ameiurus nebulosus*), a species introduced less than two centuries ago. Under laboratory conditions, we tested whether visual and chemical cues from these two predators induced different refuge-seeking and feeding behaviour responses by kōura. Predator presence produced a NCE on refuge use, with kōura exposed to catfish cues showing significantly greater refuge-seeking behaviour than those exposed to eel cues. This is consistent with the generalisation theory triggered by unfamiliar predator cues rather than classic prey naivety. Our study indicates catfish incursions may provoke a greater NCEs on kōura refuge use than resident eels. As catfish range expansion continues, these NCEs represent an ecologically significant and underappreciated threat to kōura populations, compounding the direct predation pressure already documented for this invader.

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1 Introduction

Biological invasions can have profound impacts on invaded ecosystems, particularly through the introduction of novel predators that prey upon native species lacking an evolutionary history with them (Doherty et al. 2016). Native prey may be naïve not just to a specific invader, but to entire functional groups of predators, making them especially vulnerable to direct predation when an invasion introduces a predator type entirely absent from their evolutionary history [Sih2010; Carthey2014]. In addition to these consumptive effects, native prey may exhibit heightened fear responses to invasive predators, leading to the development of anti-predator behaviours such as increased refuge use and avoidance of open habitats (Kindinger and Albins 2017). These behavioural changes, termed non-consumptive effects (NCEs), can substantially influence population dynamics by reducing foraging activity, growth, and reproductive output (Preisser et al. 2005). In contrast, native predators generally elicit different responses because prey and predators have co-evolved, resulting in behavioural and physiological adaptations that stabilise predator–prey interactions (Abrams 2000).

Freshwater crayfish often live in turbid flowing water or in the dark depths of lakes where they rely primarily on chemical cues (kairomones) rather than visual cues to detect and respond to predatory fish (Blake and Hart 1993; Brönmark and Hansson 2012; Wood and Moore 2020a). These kairomones carry signals via excreta, mucus, or skin, that crayfish use to assess predator risk, responding more strongly to predators that have consumed conspecifics and to alarm cues released when prey is injured or killed (Gherardi et al. 2011; Beattie and Moore 2018). Exposure to kairomones from predatory fish can generate NCEs, including reduced growth and reduced activity levels in crayfish (Adams 2007; Ramberg-Pihl and Yurewicz 2020), and even non-predatory fish can cause these behaviour changes, likely through heightened generalised awareness of risk (Nyström 2005). However, the strength of these responses depends on prior experience as wild-caught individual crayfish show stronger antipredator responses than farmed conspecifics naive to predators (Martin 2014), indicating that both learned behavior and evolutionary history shape kairomone detection.

In Aotearoa-New Zealand (hereafter Aotearoa), the endemic northern freshwater crayfish *Paranephrops planifrons* (White, 1842) commonly known as kōura is a taonga (‘culturally treasured’) species for Māori, the indigenous people of Aotearoa and is valued as a food resource (Kusabs and Quinn 2009). Ecologically, kōura play a critical role in freshwater ecosystems as ecosystem engineers through their effects on sediment dynamics, nutrient cycling, and benthic community structure, and have been described as keystone species in many freshwater systems (Collier et al. 1997; Parkyn et al. 2001; Usio and Townsend 2004). Currently kōura are threatened by multiple interacting stressors, including climate change, water-quality degradation and invasive species, contributing to widespread population declines (Lee et al. 2025; Kusabs et al. 2026).

Kōura have a long co-evolutionary history with the native longfin eel (*Anguilla dieffenbachii*) (hereafter eel) and shortfin eel (*A. australis*) (Brown 2009; MacNeil et al. 2025). Reflecting this history, kōura respond more strongly to eel kairomones than to those of introduced non-native brown trout (*Salmo trutta*), even when sourced from streams where both predators were present, suggesting that co-evolutionary history, rather than individual experience alone, shaped kōura predator recognition (Shave et al. 1994). One of the invasive species that threatens kōura is the non-native brown bullhead catfish (*Ameiurus nebulosus*) (hereafter catfish) which was introduced to Aotearoa in 1877 and has since spread through many waterways on the North Island (McDowall 1990; Barnes and Hicks 2003; Dedual 2019). For example, a decade ago catfish were confirmed present in two large volcanic lakes, Lakes Rotoiti and Rotorua in the Bay of Plenty region, which also contain kōura (Lee et al. 2025). The catfish is highly tolerant of a wide range of environmental conditions (Scott and Crossman 1973; Lee et al. 2025) and, due to its broad gape and opportunistic feeding behaviour, is capable of consuming a wide variety of native aquatic organisms (Scott and Crossman 1973; Dedual 2019). In Aotearoa, which originally has no native fish remotely similar to a catfish the recent and ongoing spread of catfish into North Island waterways represents a truly novel predation threat to which kōura may lack adequate behavioural or sensory adaptations.

In this study, controlled laboratory experiments were used to test whether novel vs native

predator presence and associated chemical cues induced changes in NCEs (refuge-seeking and feeding behaviour) by kōura. Our experiments were designed to test which of two competing hypotheses best describes kōura behavioural responses to native and invasive predator cues. 1. Kōura would show more shelter-seeking behaviour when presented with eels compared to catfish, due to naivety of the kōura towards a novel predator such as catfish (Carthey and Banks 2014). Or 2, kōura would respond more strongly to unfamiliar catfish cues than to the co-evolved eel, consistent with a generalised fear response to a novel predator type (Sih et al. 2023). Further, we expected that light regime would have an influence on shelter seeking behaviour of kōura, which like many crayfish, are most active nocturnally (Devcich 1979; Johnson et al. 2014). We also predicted that kōura feeding would be more inhibited when exposed to eel kairomones than to those of the catfish, reflecting the same co-evolutionary recognition expected to drive greater refuge use in response to eels. The overall aim of the refuge and feeding behaviour studies were to contribute to understanding the behavioural mechanisms underlying kōura vulnerability to invasive catfish and how these may differ from and be additional to, those already manifested in the presence of native predators such as eels.

2 Materials and Methods

2.1 Animal collection and preparation

In mid-August 2025, 92 kōura were sourced from two catfish-free lakes, (Ōkāreka and Tikitapu (Blue Lake)) in the Rotorua Te Arawa Lakes area, Bay of Plenty region of the North Island. Eels are present in most of the rivers and lakes of Aotearoa (McDowall 1990) and may be present in both lakes, but scarce. In late September 2025, catfish and eels were sourced from Lake Rotorua in Hamilton in the Waikato region of the North Island. Experiments were conducted at the Aquatic Research Centre of the University of Waikato. All three species were held separately in holding tanks and acclimated for a week in dechlorinated tap water under the following controlled regime: a 10:14 light: dark cycle reflecting the correct seasonal day-length at a stable temperature of 15.6 °C, pH 8.4 and 10 mg l⁻¹ dissolved oxygen, the latter parameters reflecting typical physiochemical condition in source lakes. Kōura and fish were fed every other day on Hikari Tropical carnivore fish food pellets (animal protein 47%). Because kōura are taonga (culturally treasured) species, permission had to be sought from relevant iwi (Māori tribal unit), as well as relevant government. All experimental procedures were conducted under University of Waikato animal ethics committee approval (protocol number: 1232; see Ethical Approval section).

2.2 Refuge experiment

Prior to the experiments all animals were individually weighed and measured. Only male kōura were used as male crustaceans are usually the most bold and aggressive (Stein and Magnuson

1976; Figler et al. 2005) and restricting test animals to a single sex allowed a smaller overall number of kōura to be used. Kōura had a mean ($\pm$ SD) orbital carapace length (OCL) of 26.0 mm ($\pm$ 3.08) (18.35–32.95 mm range) and a mean ($\pm$ SD) wet weight of 11.7 g ($\pm$ 4.25) (3–23 g range). Catfish ranged in length from 260–267 mm and 2.34–3.16 kg wet weight and eels ranged in wet weight from 2.50–3.52 kg, both these size ranges being able to consume adult kōura (Hicks 1997; Barnes and Hicks 2003; Ryan 2009).

Experiments were conducted in 38 L glass tanks (600 $\times$ 200 $\times$ 220 mm) with each tank having three sides (apart from front of tank) blacked out to block visual cues from neighbouring tanks. Each tank was divided into two compartments by a 0.5 mm steel mesh divider, allowing visual and chemical cues between compartments, while preventing any physical contact between fish and kōura on either side of the barrier (Figure 1). All tanks were filled with ~26 L dechlorinated tap water and had identical environmental regimes as previously detailed for acclimatisation tanks, with water flow provided by an air stone. Each tank housed a single kōura on one side of the barrier and either a single predator (catfish or eel) or no predator (control) on the other side. A halved terracotta pot (90 mm diameter $\times$ 90 mm depth) was positioned on the kōura side of the tank to serve as a potential refuge. To control for potential tank location or compartment side effects, the position of predator treatments within the aquaria was systematically rotated between experimental rounds. The experiment was repeated across five rounds, resulting in 11 catfish replicates, 11 eel replicates, and 10 control replicates and was first carried out in the light (~400 lux; n = 32) before being repeated in the dark (~70 lux; n = 32). Observations were made in the dark using red lighting positioned centrally above the tanks to minimize disturbance, as many aquatic species have limited sensitivity to red wavelengths and are unlikely to perceive this light. Following each individual experimental run, each tank was thoroughly cleaned and refilled with dechlorinated tap water before reuse. Each individual kōura was only exposed once to a fish predator and there were no mortalities of kōura or fish during experiments. All experiments were conducted within a three-week period during October 2025.

At the beginning of each experiment, an individual kōura was placed in the middle of the tank compartment containing the refuge and observations of each kōura location were recorded at 21 time intervals within a 5 hour period (the intervals being: 0, 5, 10, 20, 30, 45, 60, 61, 65, 70, 80, 90, 120, 150, 210, 240, 245, 260, 275, 290, and 300 minutes). The 0-minute recording was the individual's initial immediate location after its addition to the tank. Kōura location was recorded as either refuge (the animal was using the refuge, being inside it or immediately behind it and touching it) or non-refuge/ in the open (any other location in the tank where the animal was not in the refuge or touching the refuge). The fish predator was introduced into the tank on the other side of the barrier from the kōura compartment, after 60 minutes and removed after 240 minutes. The compressed intervals around 60–65 minutes and 240–245 minutes were designed to capture immediate behavioural responses at the point of predator introduction and removal respectively. The control tanks had no fish added but at 60 minutes and again at 240 minutes, the water in the 'predator' compartment was mechanically disturbed by agitating the water with a small hand net for approximately 30 seconds. This was to simulate the disturbance time associated with adding and removing fish in the other treatments

with the same size hand net. At the end of each experiment, the kōura was removed after the final 300-minute observation.

To analyse kōura refuge use, a beta-binomial generalized linear mixed-effects model (GLMM) was fitted using the `glmmTMB` package in R (Brooks et al. 2017; R Core Team 2025). To address potential non-independence of repeated observations within tanks, raw observations were first aggregated into three experimental periods (before, during, and after predator introduction) per individual kōura, with the number of observations in refuge out of total observations per period used as the response variable. Treatment (catfish, eel, control), light condition (light, dark), and period (before, during, after) were included as fixed effects. A beta-binomial distribution was used to account for overdispersion in the binomial proportion data. Nine candidate models were compared, differing in fixed effect structure (main effects only, two-way interactions, and full three-way interaction) and random effects inclusion (no random effects, tank identity, and tank plus experimental round as random intercepts), using the corrected Akaike Information Criterion (AICc), appropriate given the ratio of sample size to estimated parameters ($n/k < 40$; Burnham and Anderson (2002)). The best-supported model included treatment, light condition, and period as fixed effects with no random intercept, indicating that variance among tanks and rounds did not improve model fit beyond the fixed effect’s structure. Model fit of the final model was assessed using DHARMA simulation-based residuals, including tests for overdispersion and zero-inflation (Hartig et al. 2024). Marginal means and pairwise contrasts between factor levels were obtained using the `emmeans` package (Lenth et al. 2026). Spearman rank correlations with Bonferroni-adjusted p-values were also used to investigate potential relationships between kōura body size (OCL) and overall refuge use (all time intervals combined) in the three treatments.

2.3 Feeding experiment

Experiments were conducted in glass tanks ($320 \times 170 \times 170$ mm), containing 5 L dechlorinated tap water and a halved terracotta pot (90 mm diameter $\times$ 90 mm depth). Catfish and eels had been held at the same density in separate identical holding tanks for a week prior to experiment. Fish exudates were collected by placing four fish (catfish or eel) in a 10 L bucket filled with water from the holding tanks for 15 minutes to concentrate the kairomones in the water. For eel and catfish treatments, 1 L from the buckets was added to each tank already holding 5 L of tap water, and for control an additional 1 L of dechlorinated tap water was added. Next, 10 Hikari Tropical carnivore pellets were added to each tank. Trial experiments showed these pellets remained intact for over 24 hours in tap water. In three rounds, a total of 46 male kōura were measured and weighed. Their mean ($\pm$ SD) OCL was 27.93 mm ($\pm$ 4.80), with a range of 20.48 to 40.88 mm, and the mean ($\pm$ SD) wet weight of 15.83 g ($\pm$ 8.99), ranging from 6.00 to 42.00 g. Each kōura was then placed individually in a tank. For each round kōura were starved for different periods (either 0, 3 or 4 days) prior to the feeding experiment to also investigate how starvation period influenced feeding motivation. After 0 days starvation, 14 kōura were exposed to eel water ($n = 4$), catfish water ($n = 6$) or control water ($n = 4$). After

3 days starvation, 16 kōura were exposed to eel water ($n = 6$), catfish water ($n = 5$) or control water ($n = 5$). After 4 days starvation, 16 kōura were exposed to eel water ($n = 5$), catfish water ($n = 5$) or control water ($n = 6$). After 24 hours, when each experiment was terminated, the number of pellets eaten (to the nearest quarter pellet) was counted and the feeding rate (pellets eaten per gram of wet weight (WW) of kōura) calculated. Kōura feeding rate was analysed using a linear model with pellets consumed per gram wet weight as the response, with treatment (catfish, eel, control) and starvation period (0, 3, 4) as fixed effects. Because each starvation period was conducted in a separate experimental round, starvation period and round were fully confounded by design, precluding the inclusion of round as a random effect. Overall significance of fixed effects was assessed using Type III ANOVA (car package; Fox and Weisberg (2018)), and model assumptions were verified using residual plots and testing normality of residuals using the Shapiro-Wilk test. Pairwise comparisons between treatment and starvation level combinations were conducted using estimated marginal means.

3 Results

3.1 Refuge experiment

The proportion of time kōura spent in refuge varied across treatments, experimental periods, and light conditions (Figure 2). Stratified analyses revealed that treatment effects on refuge use differed between light and dark conditions. Under light conditions, there were no significant differences in refuge use among the three treatments (Table 1). In contrast, under dark condition, kōura exposed to catfish used refuge significantly more than those exposed to eels ($p = 0.004$), with control treatment refuge use was intermediate ($p = 0.083$ vs catfish, $p = 0.565$ vs eel). Regarding experimental periods, refuge use differed significantly between periods in light conditions, with lower refuge use before predator introduction compared to after predator removal ($p = 0.021$). No significant period effects were observed in dark conditions. The combined model showed that light regime had a significant effect on refuge use ($p = 0.005$), with kōura using refuge more in light than in dark conditions across all treatments. Overall, under light conditions, the pattern of refuge use in the control treatment was more similar to the catfish treatment, whereas under dark conditions it more resembled the eel treatment (Fig. 3; Table S1). There was also no relationship between kōura OCL and refuge use in any of the three treatments ($r_s = r_s = -0.09$, $p = 0.693$, $r_s = 0.09$, $p = 0.680$ and $r_s = 0.16$, $p = 0.494$ for catfish, eel and control treatments respectively, all NS).

For all treatments individual kōura were highly variable in their movement around tanks and use of refuges, even in the time intervals comprising the first 60 minutes of the experiment, before fish were introduced. The coefficient of variations (CVs) for kōura refuge use in the light and the dark before fish introduction were similarly high, being 60.4 % and 58.2 % (both $n = 32$) respectively, confirming high baseline variation in kōura refuge use, regardless of treatment type. Despite this inherent variation in individual kōura behaviour, clear patterns in kōura refuge use were evident over the course of the experiment. Under light conditions, the control

showed the largest overall increase in refuge use (40% to 82%), driven primarily by a sharp rise in the ‘after’ period, while the catfish treatment showed a more sustained increase across all periods (52% to 73%) and the eel treatment the smallest increase (35% to 53% Figure 3). Under dark conditions, the pattern shifted, with the catfish treatment showing the greatest increase (39% to 70%), while the eel treatment showed a modest and non-sustained increase (27% to 41% during exposure, returning to 36% after), and the control remained comparatively low throughout (37% to 47%).

3.2 Feeding experiment

The linear mixed model showed that starvation period had a significant effect on kōura feeding rate (3 days: $p = <0.001$; 4 days: $p = 0.046$; Figure 4; Table 3). Treatment was not included in the main model following model selection. Stratified analyses revealed no significant treatment effects across starvation periods. Although residuals at day 0 violated normality (Shapiro-Wilk $p = 0.0112$), the non-parametric Kruskal-Wallis test confirmed no treatment effect ($\chi^2 = 2.3011$, $p = 0.3165$). Days 3 and 4 met normality assumptions and showed no significant treatment effects ($p = 0.1495$ and 0.1573 respectively). After 4 days starvation, the mean ($\pm$ SE) kōura feeding rate (pellets g WW⁻¹) for the control was over two and a half times that of the catfish and eel treatments (0.148 ± 0.035 compared to 0.059 ± 0.039 and 0.052 ± 0.039 respectively).

4 Discussion

Prey modify their behaviour in response to predation risk, creating a ‘landscape’ of fear that affects foraging, movement, and habitat use independently of actual predation events (Laundré et al. 2001). However, until relatively recently, most ecological models still assumed that predator and prey populations interacted solely through predator-prey densities by killing and consuming prey despite it being evident that by forcing prey to adopt energetically costly defensive strategies, predators can also reduce prey densities through NCEs (Preisser and Bolnick 2008). Our study joins a number of recent studies (ie., (Sniegula et al. 2025)) showing invasive predators can generate stronger NCEs on native prey than native predators, which consequently have more profound far-reaching impacts on invaded ecosystems. Our study suggests the introduction of a novel predator effectively changes the prey’s fear landscape in ways that a co-evolved predator does not. Kōura are a keystone species in Aotearoa and an ecosystem engineer and so when a predator alters kōura behaviour, this in turn could affect resources or other species through non-consumptive pathways (Peacor and Werner 2001). Such trait-mediated indirect effects represent a form of behavioural trophic cascade (Schmitz et al. 1997), whereby predator presence alone drives ecosystem-level change. Our results also highlight that prey naivety does not always show a failure to respond but can instead produce a generalised over-response consistent with the generalisation theory (Sih et al. 2010; Ehrlman et al. 2019), which is important for predicting the ecological impacts of invasive species.

4.1 Refuge use

Kōura demonstrated greater refuge-seeking behaviour when exposed to the novel catfish than to native eel. This result is notable because it contradicts the classic prey naivety hypothesis, which predicts that prey fail to respond to novel predators (Carthey and Banks 2014). Over-responses to novel predators are less commonly reported but are consistent with fear generalisation theory, whereby prior experience with native predators primes a generalised heightened vigilance response to any unfamiliar threat (Ehlman et al. 2019; Sih et al. 2023). For naïve prey, unfamiliar cues could trigger such a response because the consequences of ignoring a potential predation threat could be fatal, even when, in reality, the actual predation risk is absent (Dawkins and Krebs 1979; Lima and Dill 1990). Such generalised responses to novel stimuli tend to be overcautious and energetically costly and are characteristic of prey with no evolutionary history with the encountered predator (Sih et al. 2010). Unlike classic prey naivety scenarios where the novel predator poses no actual threat, catfish genuinely predate kōura, meaning the heightened refuge response carries a real energetic cost but is not disproportionate to actual risk. Elevated refuge use is itself a meaningful NCE, reducing foraging time with potential consequences for growth, reproduction, and population persistence (Preisser et al. 2005; Sheriff et al. 2020). However, this behaviour may simultaneously confer protection against direct predation; experimental evidence shows that shelter significantly reduces catfish predation on crayfish (Adams 2007; Zhao et al. 2024), suggesting that enhancing or restoring habitat complexity in catfish-invaded lakes could mitigate both direct predation and fear-driven reductions in foraging, though direct field testing is needed.

Before treatments were added, refuge use was similar across treatments but highly variable, reflecting substantial individual behavioural variation. The control treatment showed increasing refuge use over time, particularly under light conditions, which is unsurprising given that kōura are nocturnally adapted and need to remain hidden during daylight to avoid visual predators such as birds (Devcich 1979). While neither predator treatment differed significantly from the control, this likely reflects the influence of light regime on baseline refuge use rather than a true absence of NCEs; under light conditions control refuge use approached that of the catfish treatment, while under dark conditions it more closely resembled the eel treatment, effectively inflating variance around the control mean. Despite this, the consistent pattern of catfish provoking greater refuge use than eels under both light and dark conditions supports a genuine difference in NCEs between novel and native predator exposure.

In contrast to catfish exposure, kōura reduced their refuge use when exposed to eels. This likely reflects the co-evolutionary history the two species share, as kōura are able to recognise eel chemical and visual cues and respond in an appropriate rather than precautionary way (Shave et al. 1994). Instead of hiding, kōura may rely on contextual information, such as predator size, movement patterns, or chemical concentrations, to modulate their response (Wood and Moore 2020b). This calibrated response makes ecological sense as longfin eels are the most widely distributed fish in Aotearoa’s freshwaters (McDowall 1990), and a kōura living in constant fear of eels would have severely limited opportunity to forage. Co-evolutionary pressure may

therefore have favoured a nuanced, context-dependent response to eels rather than a blanket avoidance strategy. This pattern was most pronounced in the dark conditions, where all three treatments began similarly but changed progressively over the course of the experiment. The catfish treatment steadily increased in refuge use, while eel treatment stayed alongside the control, declining to baseline levels by the end of the during period. This suggests that under conditions most relevant to kōura natural activity, chemical recognition of the familiar eel predator is sufficient to enable a calibrated, low-cost response, while the unfamiliar catfish continues to elicit sustained precautionary refuge use (Shave et al. 1994; Wood and Moore 2020a). Beyond predator recognition, the reduced refuge use observed in eel-exposed kōura may not simply reflect a lack of response, but could instead represent active escape-searching behaviour, kōura moving away from an area where they perceived themselves as vulnerable in the confined tank space, rather than retreating to a refuge that may offer limited protection against a predator capable of pursuing prey into confined spaces.

4.2 Feeding

NCEs are not a single process but emerge from the interaction between sensory ecology and environmental context, and the absence of a kairomone effect on feeding in this study likely reflects a combination of methodological constraints rather than an absence of NCEs on foraging behaviour. The feeding experiment which unlike the refuge-use experiment, relied purely on non-visual cues, did not demonstrate a clear fish predator NCE on kōura. This may have been due to the concentration of predator kairomones not being high enough to exceed detection thresholds (Brown et al. 2004), the relatively short pre-experiment starvation period for kōura and the restricted time frame of the experiment or a combination of all three factors. Kōura can readily detect necrotic fish tissue and readily leave refuges to scavenge on catfish, eel, and marine fish carcasses with little discrimination between fish species, indicating that chemosensory responses in kōura may depend on cue type, context, and concentration rather than predator identity alone (MacNeil et al. 2025). The concentration of live fish exudates in our experiment relative to natural conditions is difficult to determine, while kairomones were concentrated by confining fish in a bucket, in our experimental tanks they were subsequently diluted in 5 L of water and not replenished over 24 hours. Whether the resulting concentration reflects, exceeds or falls below ecologically relevant levels in lakes where kōura and predators coexist remains unclear, and future studies should attempt to quantify natural kairomone concentrations to better calibrate experimental designs (Achtymichuk et al. 2025). In addition, many crayfishes show a stronger response to alarm cues of conspecifics (Stebbing et al. 2010; Gherardi et al. 2011; Ramberg-Pihl and Yurewicz 2020), and to kairomones from predators on a conspecifics diet (Beattie and Moore 2018). Fish predators in our experiments were kept for a prolonged period on the same diet as kōura, which may have further diminished or blunted the threatening nature of signals received from kairomones (Smee and Weissburg 2006). Furthermore, the pre-experiment starvation period for kōura may have been insufficient to elicit a very strong feeding response, as crayfish such as kōura are opportunistic foragers which are necessarily adapted to going for prolonged periods without feeding (Hazlett 2003). The length

of time since feeding can influence foraging behaviour, with previous studies indicating that crayfish will be more active foragers after being starved for at least three days (Hazlett 2003). In the current study, the longest starvation period was four days and at this point there was a clear discrepancy between the control feeding rate and the much lower feeding rates in the two fish treatments. Allowing for ethics and welfare concerns, if we had extended the starvation period, this may have elicited a stronger and prolonged response. Additionally, the stress of confinement in an artificial setting may itself have altered normal kōura feeding responses, as elevated baseline stress hormones are known to modify foraging behaviour independently of predator cue exposure (Clinchy et al. 2012). Finally, feeding was measured as a single endpoint after 24 hours, providing no information on whether kōura initially suppressed feeding after kairomone exposure before eventually resuming normal feeding activity (Wright et al. (2010)). Further experiments using higher kairomone concentrations, longer starvation periods, and continuous feeding monitoring would help clarify whether non-visual predator cues generate NCEs on kōura foraging behaviour.

4.3 Conclusions

As ecosystems become increasingly novel under global change, the reliability of historical predator–prey cues may decline, amplifying behavioural mismatches (Sih et al. 2011). This study provides experimental evidence that the invasive brown bullhead catfish exerts stronger NCEs on the native freshwater crayfish kōura than the co-evolved native longfin eel. The heightened refuge-seeking response to catfish likely reflects a generalised antipredator response triggered by unfamiliar chemical and visual cues, consistent with a fear generalisation framework. In contrast, the reduced response to eels suggests that co-evolutionary experience enables kōura to make more calibrated risk assessments. Comparative studies of kōura sourced from catfish-invaded versus catfish-free catchments would be worthwhile to assess whether repeated exposure over multiple generations has resulted in rapid evolution or behavioural plasticity in antipredator responses, and whether catfish-experienced kōura show more calibrated responses to catfish presence than naïve kōura. Such evolutionary dynamics could have important implications for predicting kōura persistence in catfish-invaded systems (Sih et al. 2010). From a management perspective, translocation of kōura from catfish-experienced source populations into catchments facing imminent catfish invasion may represent a more feasible strategy for seeding populations with individuals capable of more calibrated responses. Although we found no significant NCE of fish exudates on feeding behaviour, our study suggests further investigations of non-visual predator cues on kōura foraging behaviour is warranted. Because kōura are keystone species and ecosystem engineers, even a small increase in refuge use driven by catfish presence could have cascading effects on sediment dynamics, decomposition, and nutrient cycling in invaded lakes. As brown bullhead catfish continue to spread across Aotearoa’s rivers and lakes, management strategies aimed at protecting kōura should account for the behavioural cost of catfish NCEs alongside direct predation pressure as an underappreciated and ecologically significant dimension to this invasion.

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5.2 Authors’ contributions

Study conception and design were undertaken by O.V.R, C.M. and R.H. Experiments were executed by O.V.R. and C.M. Data analysis was performed by O.V.R. and F.L. Local ecological knowledge kōura were supplied by I.A.K.K. Manuscript drafting was led by O.V.R. and C.M. All authors contributed to manuscript revision, approved the results, and the final manuscript.

5.3 Ethical approval

This study received ethical approval by University of Waikato animal ethics committee, protocol number: 1232. Additional permission to conduct experiments with kōura was granted by Te Arawa Lakes Trust and Te Komiti Whakahaere.

5.4 Declarations

The corresponding author has declared that none of the authors has any competing interests.

5.5 Data availability statement

The code and derived data supporting the findings of this study are openly available on GitHub (https://github.com/OlivierRaven/Koura_Fear) and a rendered version of the analysis is hosted at https://olivierraven.github.io/Koura_Fear/. The primary raw dataset is archived on Zenodo (<https://doi.org/10.5281/zenodo.19716761>). Peer review documents, including the cover letter and reviewer responses, are available in the manuscript repository in the interest of open and transparent science.

6 Tables

Table 1: Pairwise contrasts for kōura refuge use in light and dark conditions. Results are shown separately for each light condition from beta-binomial GLMMs. Odds ratios and p-values are shown for treatment and period comparisons. P-values are Sidak-adjusted for multiple comparisons within each factor.

Condition	Contrast	Odds ratio	SE	df	Null	z value	p value
Light	control / catfish	0.863	0.396	Inf	1	-0.322	0.944
Light	control / eel	1.641	0.734	Inf	1	1.108	0.509
Light	catfish / eel	1.903	0.866	Inf	1	1.414	0.333
Light	before / during	0.463	0.194	Inf	1	-1.836	0.158
Light	before / after	0.297	0.135	Inf	1	-2.672	0.021
Light	during / after	0.641	0.294	Inf	1	-0.969	0.597
Dark	control / catfish	0.393	0.172	Inf	1	-2.133	0.083
Dark	control / eel	1.542	0.655	Inf	1	1.019	0.565
Dark	catfish / eel	3.924	1.693	Inf	1	3.170	0.004
Dark	before / during	0.728	0.286	Inf	1	-0.807	0.698
Dark	before / after	0.527	0.215	Inf	1	-1.569	0.259
Dark	during / after	0.723	0.290	Inf	1	-0.809	0.698

Source: [Analysis Notebook](#)

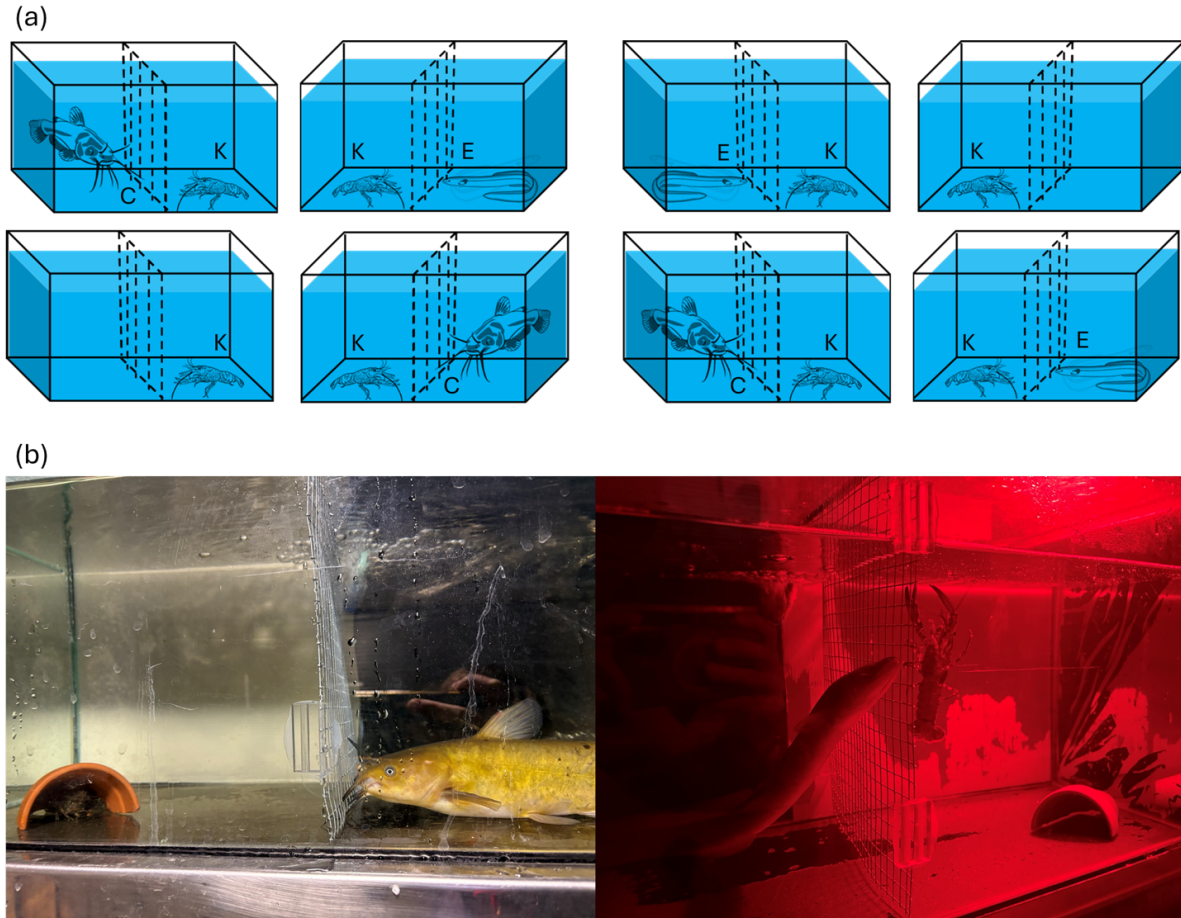


Figure 1: (a) Aquarium layout for kōura refuge seeking behaviour experiment. Each experimental round consisted of eight aquaria run simultaneously. Each tank contained a single kōura (K) and refuge (half terracotta plant pot) on one side of a mesh divider. On the other side was either a single catfish (C), eel (E) or a control with no fish. The mesh divider allowed visual and chemical cues while preventing direct physical contact between tank occupants. (b) Experiments were repeated in light and dark (red light), with photographs showing a catfish in the light regime (left) and an eel in the dark regime (red light) (right) separated from kōura by mesh dividers.

7 Figures

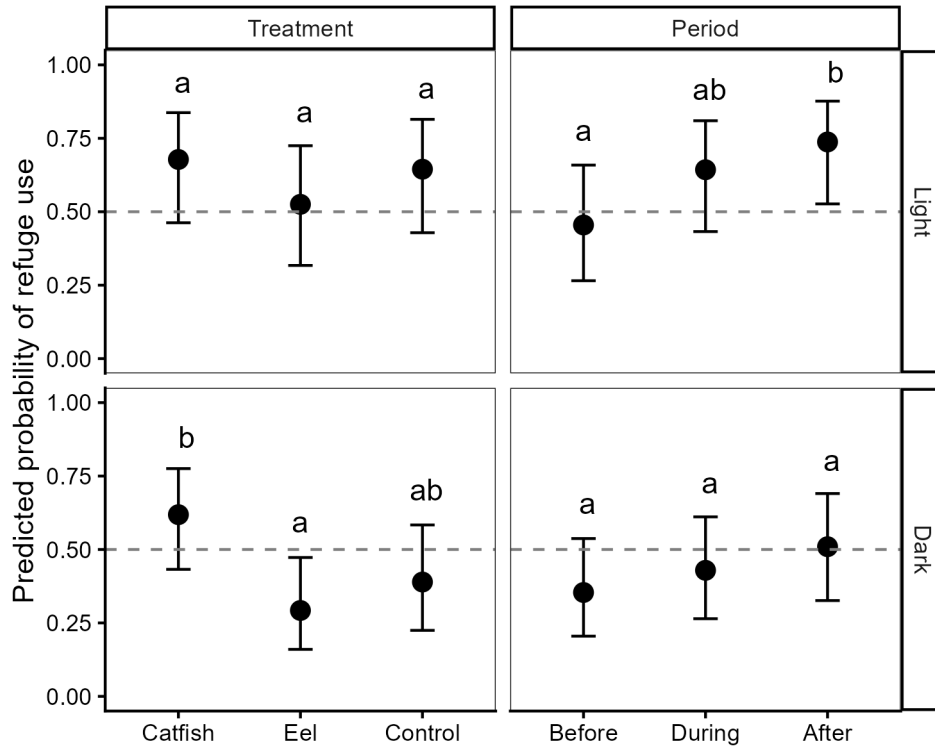


Figure 2: Predicted marginal probabilities of refuge use by treatment and experimental period in light and dark conditions. Results are shown separately for light (top) and dark (bottom) conditions, each fitted with a beta-binomial GLMM. Points represent model-estimated marginal means with 95% confidence intervals shown as error bars. Different letters (a, b) indicate statistically significant differences based on Tukey-adjusted pairwise comparisons within each factor. The dashed line at 0.5 marks the threshold where kōura spend half their time in refuge.

Source: [Article Notebook](#)

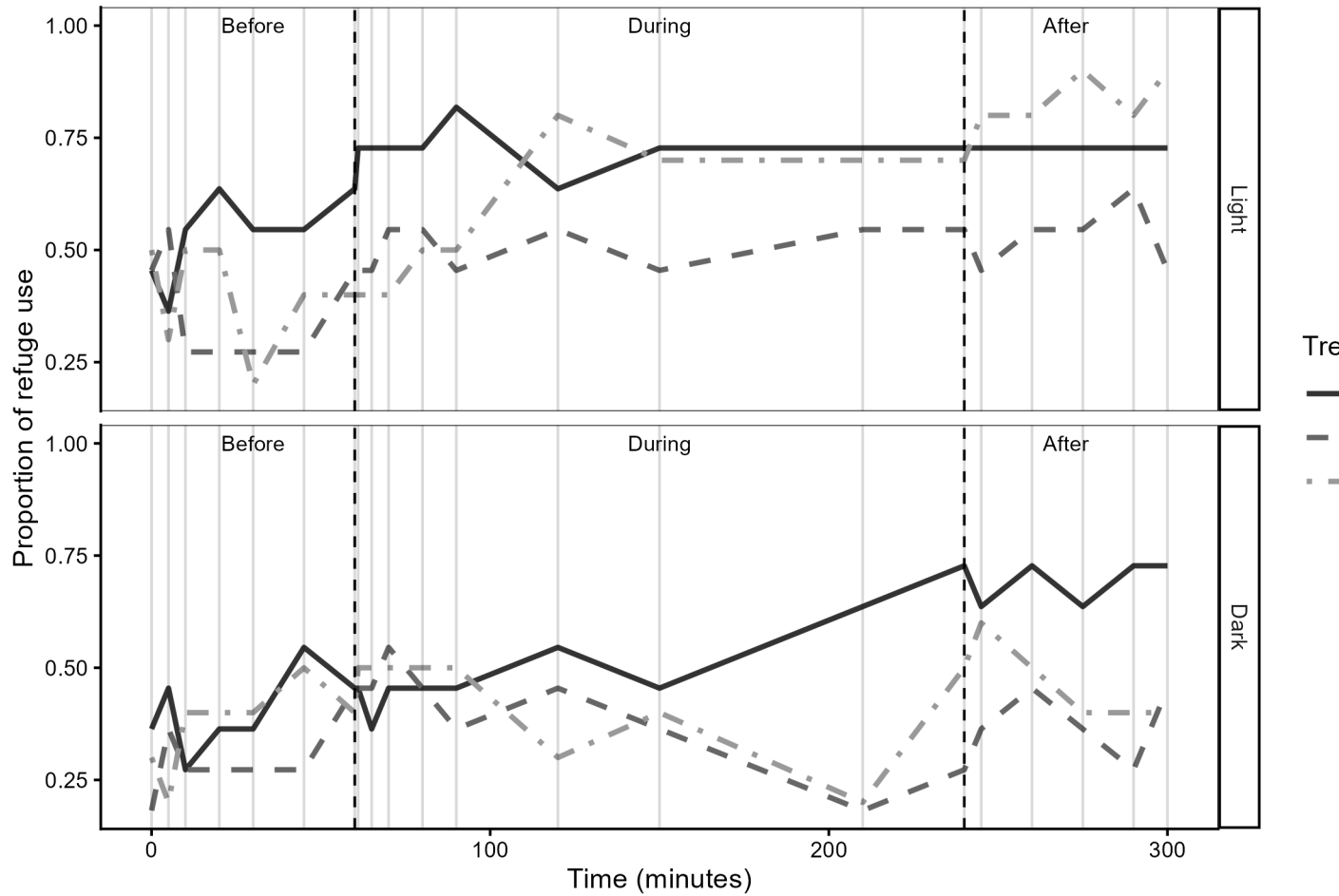


Figure 3: Kōura refuge use over the course of the experiment, showing the proportion of time spent in refuge across three periods (before, during, and after predator exposure) for each treatment (catfish: dark grey solid line; eel: medium grey dashed line; control: light grey dot-dash line) under light (top) and dark (bottom) conditions. Black dashed vertical lines indicate the start and end of predator exposure and grey vertical lines show individual observation times.

Source: [Article Notebook](#)

Table 2: Proportion of kōura observations recorded in refuge across experimental treatments (catfish, eel, control), light conditions (light, dark), and periods (before, during, and after predator exposure). Combined rows summarise across all levels of a given factor. Values represent the proportion of total observations where kōura were located in the refuge.

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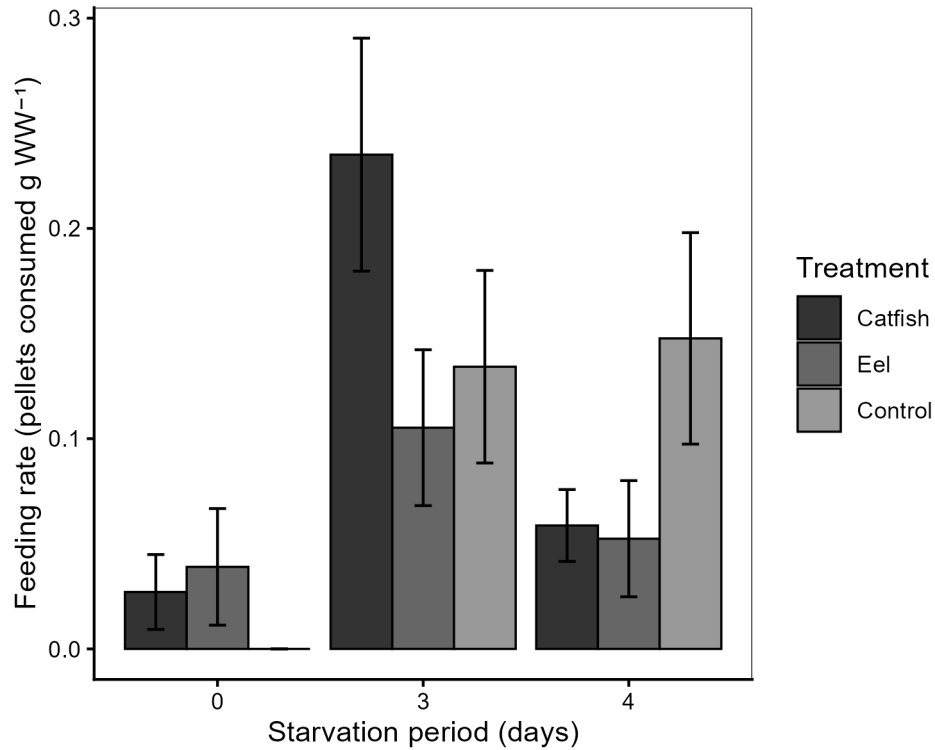


Figure 4: Mean ($\pm$ SE) kōura feeding rate (pellets consumed per gram wet weight kōura) across predator treatments (catfish, eel, control) and starvation periods (0, 3, and 4 days).

Source: [Article Notebook](#)

8 Supplementary information

Source: [Analysis Notebook](#)

Table 3: Fixed effects from the linear mixed model of kōura feeding rate (pellets consumed per gram body weight). Starvation period included as fixed effect with tank as random intercept.

Term	Estimate	Std. Error	df	t value	p value
(Intercept)	0.023	0.024	43	0.951	0.347
day_starve3	0.132	0.033	43	4.033	0.000
day_starve4	0.067	0.033	43	2.056	0.046

Source: [Analysis Notebook](#)

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